EX.No.2.1 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

AIM:

To create a 3D model from the given isometric drawing using fusion 360 tool.

REQUIREMENTS:

1.Autodesk fusion 360
2.Windows 11
3.16 GB of RAM (integrated graphics recommend 1 GB or more)
4.2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

STEP-1:

Start the sketch command
Select Sketch > Create Sketch.

STEP-2:

Select the Sketch Plane
Select the Front (XZ) Plane

STEP-3:

Create a Rectangle
Select Sketch > Rectangle > Center Rectangle.
Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.
Click once to begin placing the rectangle.
Type the Tab button and enter the values 100 mm and 60mm for the width and height respectively.
Click again to place the rectangle.
Click Finish Sketch after completing.

STEP-4:

Extrude Rectangle in XZ Plane
Select Extrude.
Select the Rectangle sketch.
Click again to place the value.
Type in the value of 30mm.
Press ENTER to accept the value.

STEP-5:

Start the sketch command
Select Sketch > Create Sketch.

STEP-6:

Select the Sketch Plane .Select the FRONT SURFACE of rectangular block

STEP-7:

Create a CIRCLE
Select Sketch >CIRCLE > CENTER CIRCLE.
Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.
Click once to begin placing the CIRCLE.
Type the Tab button and enter the value DIA 100mm .
Click again to place the circle.
Click Finish Sketch after completing.

STEP-8:

Extrude CIRCLE
Select Extrude.
Select the Rectangle sketch.
Click again to place the value.
Type in the value of 20mm.
Press ENTER to accept the value.

STEP-9:

Start the sketch command
Select Sketch > Create Sketch.

STEP-10:

Select the Sketch Plane

Select the TOP SURFACE OF RECTANGLE.

STEP-11:

Create a line

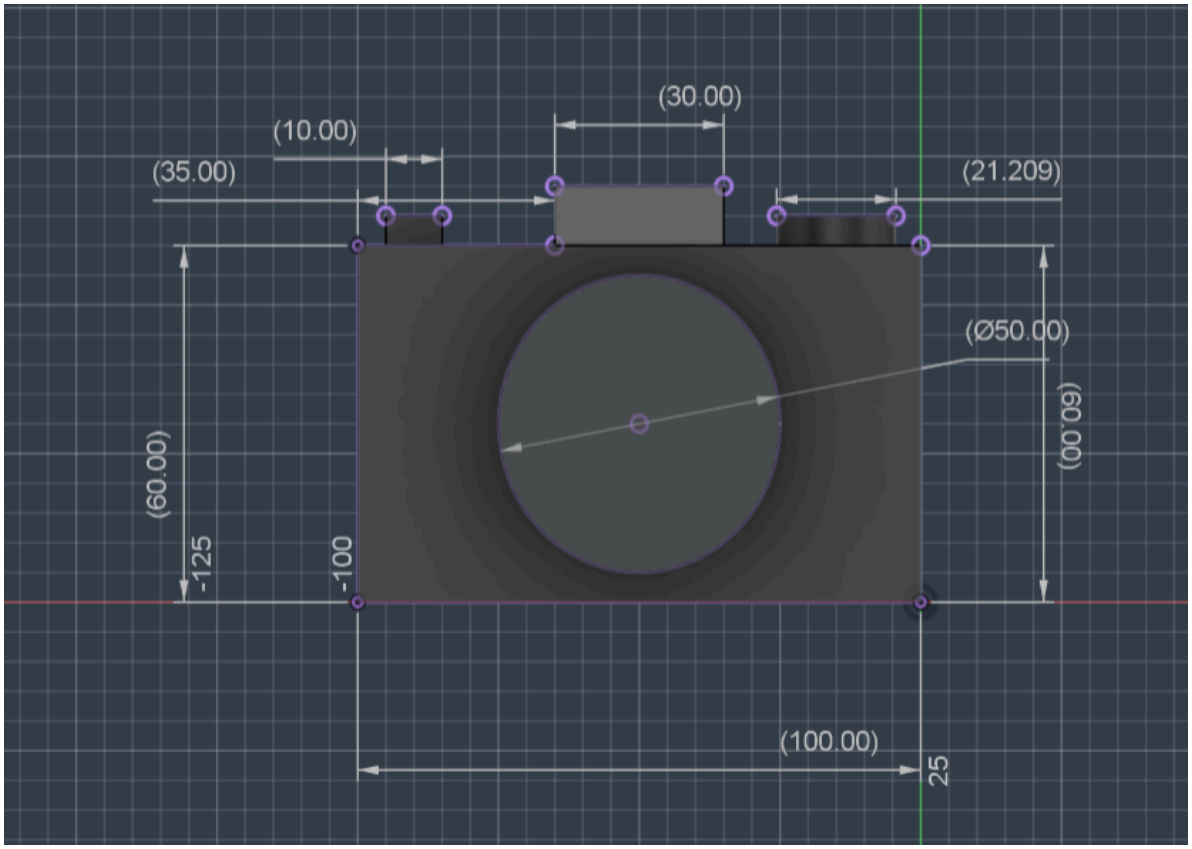
1.Select Line command.

2. Draw the projections in the diagram

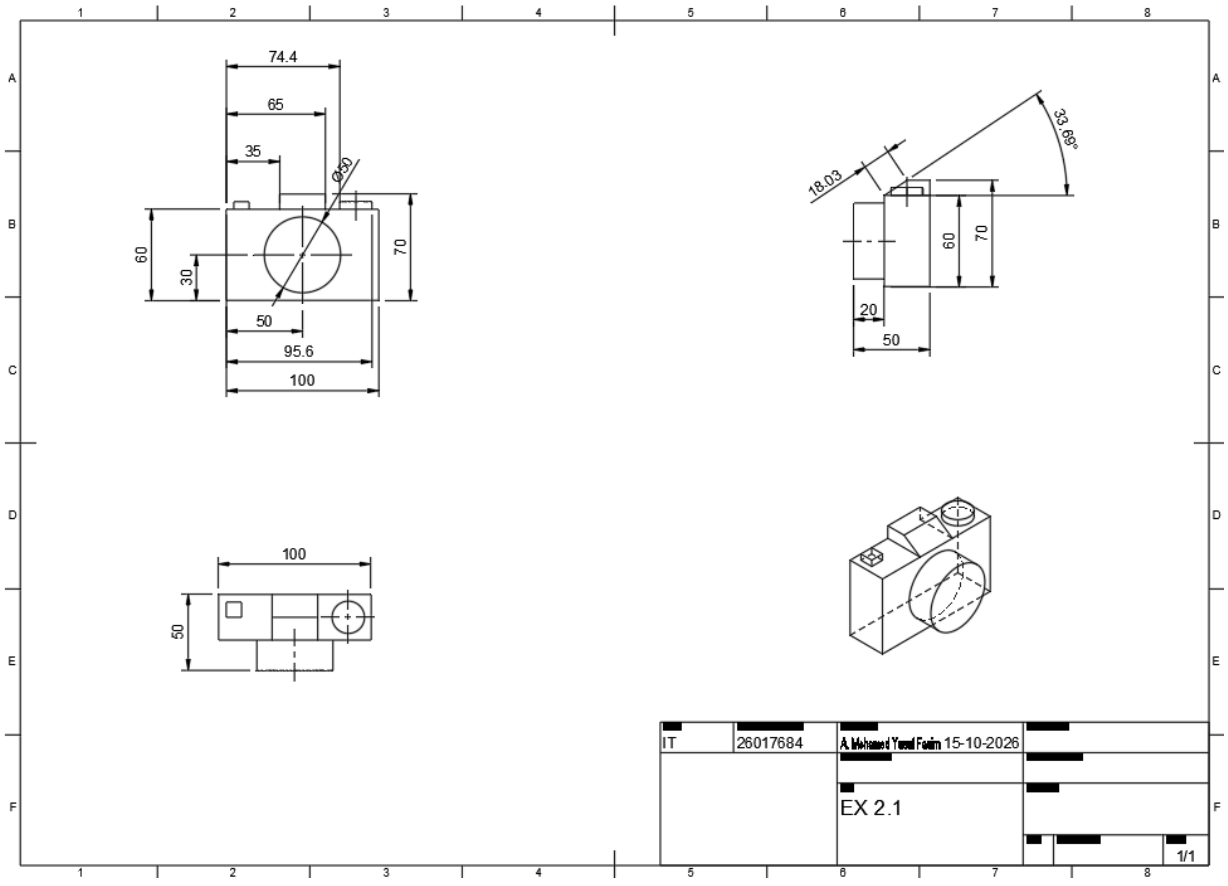
3. Click Finish Sketch.

4. extrude the sketch.

SKETCH:



OUTPUT:



RESULT:

Thus, the 3D model is created from a given isometric drawing.